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Master's Degree in Artificial Intelligence and Data
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Intelligent Monitoring of Escape Routes via Computer Vision

Advisors:

Eng. Giustino Claudio
Miglionico

Prof. Francesco Marcelloni

Prof. Pietro Ducange

Author:

Andrea Innocenti

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Abstract

[Insert thesis abstract in English here. This should provide a concise summary of the research problem, methodology, key results, and conclusions - typically 150-300 words.]

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Chapter 1

Introduction

1.1 Motivation

[Contextualize the problem and justify the work]

1.2 Research Question

[Clear statement of the research question]

1.3 Thesis Outline

[Overview of remaining chapters]

Chapter 2

State of the Art

2.1 Vision-Based Safety Monitoring in Built Environments

Safety inspections of buildings and workplaces have traditionally been carried out by human operators who periodically walk the premises, identify hazards, and verify compliance with safety regulations. The literature consistently identifies the same limitations of this manual process: it is costly, it scales poorly with the size of the monitored facility, and its low frequency leaves long blind intervals during which a hazard can persist undetected [7, 3]. Computer vision offers a natural path to automation, since most safety-relevant conditions - a blocked corridor, a missing helmet, a misplaced object - are directly observable in images, and camera infrastructure is comparatively cheap to deploy and maintain.

The condition of evacuation routes is a recurring subject in this literature. Er-can and Mathyalakan [7] address clutter in the common areas of Singapore public housing, where residents' belongings - bicycles, laundry racks, shoe racks, plants - accumulate along corridors and may hinder emergency response; safety guidelines mandate a clear escape passage, but enforcement through on-site inspection is infrequent and labour-intensive. Their system detects clutter with YOLO-family object detectors mounted on an autonomous mobile robot and reports violations, with location and type of obstruction, to a remote database. Along the same line, fire-safety regulations are the explicit target of Chen et al. [3], who observe that automated compliance checking has historically been confined to the *design* phase, operating on structured building models, and therefore cannot capture hazards that arise only during building *operation* - obstructed exits, relocated equipment, informal modifications of the space. Their proposed remedy combines deep-learning perception modules with a knowledge graph encoding spatial relationships and regulatory rules, so that raw on-site imagery can be turned into structured, regulation-aware compliance reports.

A third strand concerns construction sites, arguably the most mature application domain for vision-based safety monitoring. Beyond detecting unsafe configurations

of equipment and personnel, recent work has concentrated on the *reliability* of the monitoring signal itself: Zaidi et al. [23] note that systems classifying each video frame in isolation inevitably produce false alarms whenever a single frame is mis-detected, and introduce a temporal-analysis module that aggregates evidence over a window of frames before raising an alarm. The concern they articulate - that a monitoring system crying wolf is quickly ignored by its operators - is general, and returns in Section 2.5 as a central design constraint for the present work.

Across these systems three common traits emerge. First, the sensing infrastructure consists of fixed or slowly-moving cameras observing a known scene, which makes the scene layout itself prior knowledge that the algorithms can exploit. Second, the monitoring must run continuously and in (near) real-time, which constrains the admissible computational budget. Third, imagery of building interiors is privacy-sensitive, and streaming it to a remote server is often undesirable or prohibited; together with bandwidth and per-camera cost considerations, this pushes the computation toward the network edge, a constraint examined in the next section.

2.2 Object Detection on Edge Devices

The most direct way to recognise an obstruction in an image is to detect the obstructing object explicitly. Deep-learning object detectors are commonly divided into two-stage models, which first propose candidate regions and then classify them, and one-stage models, which predict classes and bounding boxes in a single forward pass [12]. The YOLO family [15] is the prototypical one-stage design and has become the de-facto standard for real-time applications: it processes the whole image with one convolutional network, trading a modest amount of accuracy for an order-of-magnitude gain in speed over two-stage models. Successive YOLO generations, together with dedicated lightweight architectures, quantisation, and inference engines such as TensorRT and OpenVINO, have progressively lowered the computational cost of detection to the point where it fits the resource envelope of single-board computers; Mittal [12] surveys this landscape and the reasons driving it - bandwidth reduction, latency, energy, and the privacy advantage of keeping image data on the device that captured it.

How far this paradigm can be pushed on low-cost hardware is well documented by the line of work on personal protective equipment (PPE) monitoring, a safety application whose deployment constraints closely mirror those of exit monitoring. Gallo et al. [8] argue against the prevailing cloud-based architectures - which require sustained bandwidth, a reliable connection, and the off-site transfer of workers' images - and deploy PPE detection entirely on a Raspberry Pi 4 with a USB neural accelerator, selecting YOLOv4-Tiny as the best accuracy/throughput compromise at about 6.8 frames per second. On the same dataset, Miglionico et al. [11] evaluate the YOLOv5 size spectrum on identical hardware, with the nano variant reaching 7.9 frames per second at 81.7% mAP@50, and later extend the comparison to YOLOv11 and the transformer-based RT-DETR on an NVIDIA Jetson Nano [10]: the nano

model with TensorRT optimisation achieves 10.6 frames per second within a 3W power budget, whereas transformer detectors, although competitive in accuracy, prove too slow for real-time edge inference. Taken together, these studies establish that modern compact detectors deliver usable accuracy at interactive frame rates on hardware costing tens to a few hundred euros - a premise this thesis builds upon.

Closer to the problem addressed here, object detectors have been applied to the monitoring of evacuation routes themselves. Besides the corridor-clutter robot of Ercan and Mathyalakan [7] discussed above, Shin et al. [19] use a YOLOv8 model on frames from pre-existing CCTV cameras to detect loads and obstacles in the corridors of complex buildings, and convert the detections, through a homography-based perspective correction, into an estimate of the *effective evacuation width* that remains available; after augmenting and re-labelling a custom dataset they report near-saturated metrics (precision 0.99, recall 1.0) in their static test setting. This confirms the feasibility of detector-based egress monitoring on existing camera infrastructure, the same deployment model assumed in this thesis.

The limits of the paradigm, however, are structural rather than incidental. First, a detector can only find what it was trained to find: the obstacle vocabulary is fixed at training time, while the space of objects that can block an exit is open-ended, so any closed-set detector leaves coverage gaps by construction. Second, the supervised regime requires a labelled dataset whose collection and annotation dominate the deployment cost, and whose biases bound the model's generalisation - Shin et al. [19] themselves must re-label and augment their data to cope with classes absent from public datasets. Third, detection accuracy degrades when objects are partially occluded or truncated, a pervasive condition in cluttered corridors; an entire research thread surveyed by Ouairhi et al. [13] is devoted to occlusion-handling strategies, and its very existence signals that the problem is far from solved. Within this thesis, the YOLO-based pipeline (P1) inherits both the strengths and these limitations; the approaches reviewed in the next two sections trade the ability to *name* the obstacle for independence from a predefined obstacle vocabulary.

2.3 Reference-Based Change Detection

PARLARE ANCHE DEL FATTO CHE AUTOENCODER E GAN NON SONO TROPPO ADATTI PER CHANGE DETECTION

When the camera is fixed, an alternative to recognising obstacles is to detect that the scene has *changed* with respect to a stored reference of its clear state. This idea descends from classical background subtraction, where a statistical background model is maintained over time and pixels deviating from it are labelled foreground. As Santana et al. [17] point out, however, conventional background models adapt continuously and eventually absorb long-standing foreground objects into the background - precisely the wrong behaviour for obstruction monitoring, where a cart abandoned in front of an exit must keep raising an alarm for as long as it stays there. Route obstruction therefore calls for comparison against a *fixed* reference of

the known-clear scene rather than against an adaptive background estimate.

Two recent works apply this reference-based paradigm directly to evacuation routes. Talebi et al. [20] propose a door-centric formulation of exit-blockage detection: instead of enumerating possible obstacles, the system monitors only the visual state of the door itself. A lightweight YOLOv11 network is trained on a single class, the door, and the bounding box it predicts at run time is compared - in geometry and overlap - against the box recorded in the unobstructed reference state; a blockage is inferred whenever the visible portion of the door deviates persistently from the reference. The authors argue that this bypasses the structural weakness of obstacle catalogues identified in Section 2.2, since any object large enough to obstruct the doorway alters the door’s visible geometry regardless of its category, and they demonstrate real-time operation on a Raspberry Pi. The trade-off is the coarseness of the signal: the method observes a single bounding box, so it cannot localise the obstruction, and changes that do not alter the door’s visible extent - or doors that the detector fails to find under unusual lighting - fall outside its reach. The door-centric baseline (P2) developed in this thesis follows this design.

Santana et al. [17] push the comparison to the pixel level with a Siamese architecture. Two weight-sharing U-Net encoders process the reference frame and the current frame, their representations are compared at multiple levels of abstraction, and a decoder produces a binary segmentation mask of the changed regions; training on the generic change-detection corpus CDNet2014 teaches the network what constitutes a meaningful change, without committing to any object vocabulary. Applied to PetrobrasROUTES, a dataset of obstructed and abandoned objects on escape routes of industrial plants, the approach proves markedly more robust to noise and illumination variation than classical background-modelling baselines, because the comparison happens in a learned feature space rather than in pixel intensities. This is the design the Siamese change-detection pipeline (P4) of this thesis builds upon.

Reference-based methods occupy an attractive middle ground for the problem at hand: they are object-agnostic, produce a spatial localisation of the change (in the Siamese variant), and exploit the fixed-camera deployment model naturally. Their residual requirements are nonetheless non-trivial: the Siamese comparator must still be *trained* to distinguish meaningful change from nuisance variation, which presupposes a labelled change-detection corpus, and every method in this family remains exposed to reference staleness - the stored clear-state image must be re-acquired whenever the normal appearance of the scene legitimately changes. The family of methods reviewed next pushes label-independence one step further, dispensing with change supervision altogether.

2.4 Unsupervised Anomaly Detection

The third family of approaches abandons supervision entirely: the system learns a model of *normality* from unlabelled normal data and flags, at test time, whatever deviates from it. This one-class formulation matches the data asymmetry of safety

monitoring - normal observations are abundant and free, anomalous ones are rare, heterogeneous, and expensive to stage - and it is the formulation adopted by the third pipeline of this thesis.

In video surveillance, anomaly detection is a mature research area, surveyed among others by Duong et al. [6]. The dominant problem setting is the detection of abnormal *human* behaviour - fighting, intrusions, abandoned luggage - with models trained on normal footage and evaluated on benchmarks such as UCSD Ped, CUHK Avenue, and ShanghaiTech. Two observations from this literature carry over directly to exit monitoring. First, anomaly is *context-dependent*: an event normal in one scene may be anomalous in another [6, 14], which implies that normality is best modelled per camera rather than globally - the design choice at the heart of the per-camera memory of Chapter 3. Second, the recurrent failure sources reported across methods - occlusion, low illumination, dynamic backgrounds and sensor noise [6] - are environmental rather than algorithmic, and any deployable system must budget for them explicitly. Patrikar and Parate [14] review the same field from the deployment side, observing that anomaly detection is a delay-sensitive application for which round-trips to a cloud back-end are ill-suited, and that the natural convergence point is anomaly detection executed on edge devices co-located with the cameras - the architecture assumed throughout this thesis.

A parallel and, for the present work, more directly relevant line of research developed in *industrial* visual inspection, where products photographed under a fixed viewpoint must be screened for defects never seen during training. The release of the MVTec AD benchmark [1] crystallised the modern formulation: only defect-free images are available for training, and methods are judged on both image-level detection and pixel-level localisation. The best-performing family of methods sidesteps representation learning altogether, describing normality directly in the feature space of a network pre-trained on natural images: SPADE [4] retrieves nearest neighbours of multi-scale features from a gallery of normal images, and PaDiM [5] fits a per-position Gaussian to the features of aligned normal samples. PatchCore [16] refined this embedding-based approach into the de-facto standard: locally aggregated patch features from a frozen backbone are stored in a memory bank, compressed by greedy coreset subsampling to a small fraction of its original size with negligible performance loss, and queried at test time by a re-weighted nearest-neighbour rule that yields both a scalar anomaly score and a spatial anomaly map. On MVTec AD, PatchCore achieves an image-level AUROC above 99%, while its training-free construction and compact memory make it computationally frugal at inference - properties that motivated its adoption, in adapted form, as the core of pipeline P3 (Chapter 3).

The transfer from industrial inspection to evacuation-route monitoring is conceptually natural: a fixed camera observing a doorway plays the role of the fixed inspection viewpoint, the clear exit is the defect-free product, and an obstruction is a structural defect of arbitrary, unforeseen appearance. The correspondence is not perfect, however. Industrial imagery is acquired under controlled, repeatable illumination, whereas a corridor is subject to daylight cycles, cast shadows, and re-

flections; the nuisance variability that industrial benchmarks deliberately suppress becomes, in this domain, the principal source of false alarms. Techniques for managing precisely this gap are the subject of the next section.

2.5 Robustness to Illumination Nuisances and False Alarms

For a monitoring system that runs unattended, the false-positive rate is not a secondary metric but the property that decides whether the system is usable at all: operators quickly learn to ignore an alarm that fires on harmless events, at which point the system provides negative value. The literature attacks the problem from two complementary directions - making the *decision* more conservative, and making the *model* more robust to the nuisance factors that trigger spurious detections.

The first direction is exemplified by Zaidi et al. [23], already encountered in Section 2.1: since per-frame classification amplifies every transient misdetection into an alarm, they accumulate evidence over a temporal window and raise an alarm only when the fraction of hazardous frames exceeds a threshold, measurably reducing false alarms during live monitoring. Temporal aggregation of this kind is largely orthogonal to the underlying detector and could be layered onto any of the pipelines considered in this thesis; it does not, however, help against nuisance factors that are *persistent* rather than transient - a cast shadow that sits on the floor for hours defeats any voting window.

Persistent nuisances dominate in the fixed-camera one-class setting, and illumination is by far the most important among them. Cao et al. [2] provide the systematic evidence: their M²AD benchmark images about a thousand objects under 120 combinations of viewpoint and illumination and shows that state-of-the-art visual anomaly detection methods, PatchCore included, degrade severely when test-time illumination deviates from the training condition. This quantifies the observation that closed Section 2.4: the implicit assumption of photometric consistency between the normal reference data and the test stream, harmless on industrial inspection benches, is the weak point of embedding-based anomaly detection in uncontrolled environments.

Among illumination effects, cast shadows are the most structured and the most deceptive, since they introduce sharp, object-like contours that low-level features cannot distinguish from physical objects. The classical remedy is to detect and remove shadows before analysis, a research line with a long history in traffic and outdoor surveillance; but removal adds a processing stage, and its own failures propagate downstream. An alternative, recently explored by Ju and Reibman [9] for camera-based action recognition, is to make the model itself shadow-tolerant by *injecting synthetic shadows into the training data*. Their study offers two findings of direct relevance here. First, shadow augmentation applied to shadow-free training data substantially improves accuracy on out-of-distribution test sets containing

real, heavy shadows - the exact configuration of a one-shot memory built from a clean reference image and tested on frames with environmental shadows. Second, generic photometric augmentation is not a substitute: globally darkening the training images *worsens* performance, confirming that robustness to localised, structured darkening requires explicitly structured augmentation rather than wider brightness ranges. The shadow and light augmentation scheme adopted in pipeline P3 (Chapter 3) follows this principle, extending it symmetrically to localised light beams and combining it with a spatial gating rule that exploits the scene geometry.

2.6 Positioning of This Work

The deployment scenario of this thesis - a fixed camera per emergency exit, processing on a low-cost edge device, alarms for arbitrary obstructions - induces five requirements against which the reviewed families can be measured:

1. **Open-set coverage:** any object blocking the route must be detectable, including categories never seen before;
2. **Label independence:** deploying on a new camera should require no labelled examples of obstructions;
3. **Spatial localisation:** the system should indicate *where* the anomaly lies, both for operator verification and for spatial filtering of known nuisance regions;
4. **Edge compatibility:** inference must run continuously on resource-constrained hardware;
5. **Nuisance robustness:** illumination changes, shadows, and activity behind glass doors must not raise alarms.

Table 2.1 summarises how the four approach families reviewed in this chapter - each instantiated by one of the pipelines compared in this thesis - relate to these requirements.

No single family dominates. Supervised detection offers semantic interpretability - it names the obstacle - and benefits from a mature edge-deployment toolchain [12, 10], but its closed vocabulary and labelling cost conflict with requirements 1 and 2. Door-centric monitoring [20] is open-set and extremely cheap, but its signal is binary and frame-level, sacrificing requirement 3. Siamese change detection [17] satisfies requirements 1 and 3, yet still depends on a labelled change corpus and on the maintenance of an up-to-date reference frame. One-class anomaly detection in the PatchCore style [16] is the only family that is simultaneously label-free, open-set, and spatially resolved, with a training-free build that suits edge deployment; its known weakness is requirement 5, since embedding-based methods presume photometric consistency between reference and test conditions [2].

Approach	Training data	Obstacle coverage	Localisation	Per-camera deployment
Object detection (P1) [8, 10, 19]	labelled boxes for each obstacle class	closed-set	bounding boxes	none (shared model)
Door-centric monitoring (P2) [20]	door-class labels only	open-set (indirect)	door box only	reference state of the door
Siamese change detection (P4) [17]	generic change-segmentation corpus	open-set	pixel mask	reference frame
One-class anomaly detection (P3) [16]	none (normal images only)	open-set	anomaly map	single reference image

Table 2.1: Approach families for exit-obstruction detection, each instantiated by one of the four pipelines of this thesis, against the requirements of the target deployment.

Two gaps in the literature emerge from this review. First, the methods closest to the target task have not met in prior work: industrial-style one-class anomaly detection has, to the best of our knowledge, not been applied to emergency-exit obstruction monitoring, where its data efficiency is most valuable but its illumination sensitivity is most exposed; conversely, the exit-monitoring systems in the literature either presuppose obstacle catalogues [7, 19] or renounce spatial localisation [20]. Second, no prior study compares representatives of the detection, change-detection, and anomaly-detection families on a common evacuation-route dataset under common edge constraints. This thesis addresses both gaps: it adapts Patch-Core to per-camera, one-shot exit monitoring - augmenting it with shadow and light augmentation and geometry-aware spatial gating to confront requirement 5 head-on (Chapter 3) - and evaluates it against the detection-based (P1), door-centric (P2), and Siamese (P4) alternatives on a shared dataset and experimental protocol (Chapter 5).

Chapter 3

Method

3.1 Objective

The goal of this work is to detect, on an edge device, when an obstacle blocks an evacuation route (a door or a corridor). Among the pipelines compared in this thesis, this chapter develops the third one, *Unsupervised Anomaly Detection with PatchCore* (P3), whose distinctive property is that it learns only from normal, unobstructed scenes and requires no labelled examples of obstructions. The following sections derive the components of the method, their mathematical formulation, and the data on which they operate; implementation choices and numerical settings are deferred to Chapter 4, and the empirical evaluation to Chapter 5.

3.2 Pipeline Overview

3.2.1 Unsupervised Anomaly Detection with PatchCore (P3)

Motivation and design rationale

The detection of emergency-exit occlusions presents a fundamental labelling bottleneck. Normal scenes - clear, unobstructed doorways and corridors - are easy to capture at scale, but the space of possible obstructions is open-ended: wheelchairs, carts, stretchers, waste bins, cardboard boxes, and any combination thereof can occupy a doorway in arbitrary configurations, orientations, and lighting conditions. Collecting and labelling a representative sample of anomalous scenes is time-consuming, and any finite labelled set risks leaving coverage gaps on obstacle types not seen during training.

P3 sidesteps this bottleneck by framing occlusion detection as *one-class anomaly detection*: the system learns only from normal scenes and, at inference time, flags any image whose feature representation deviates significantly from what was memorised as normal. This removes the need for labelled anomalous examples entirely. The choice to adapt PatchCore [16], originally proposed for industrial surface inspection,

to the emergency-exit domain is motivated by three properties that align with the deployment context:

1. **One-shot operation:** a single reference photograph of each monitored location is sufficient to build the entire anomaly model. In a real deployment, each fixed camera captures one reference frame of the clear exit; no further data collection is required.
2. **Spatial resolution:** PatchCore operates at the patch level rather than producing a single image-level embedding, preserving the spatial structure of the anomaly and enabling heatmap visualisation and spatially-aware filtering.
3. **Edge compatibility:** inference - feature extraction through a frozen backbone followed by a nearest-neighbour lookup in a compact memory - runs entirely on CPU, requires no gradient computation, and can be parallelised per camera without shared state.

Among the four pipelines, P3 is the most data-efficient: it needs only a single normal reference image and a pre-trained backbone, with no annotation effort to deploy on a new camera. The trade-off is that it cannot identify *which* obstacle is present - it only signals that something is anomalous - and it is, in principle, sensitive to any scene change, including non-obstacle changes such as lighting variations, shadows, or background movement. Managing these false-positive sources is the central methodological challenge of this chapter.

Key intuition

The core intuition is that the intermediate representations of a deep convolutional network pre-trained on a large natural-image corpus are broadly transferable: they capture generic visual patterns (textures, edges, object parts) that remain discriminative across domains, including surveillance footage of building interiors. A patch belonging to a normal scene of a fire-exit door yields a feature vector close to other normal patches in this embedding space, whereas a patch belonging to an unexpected obstacle - the wheel of a cart, the fabric of a stretcher, the surface of a box - yields a feature vector distant from all stored normal representations, hence a high anomaly score.

The memory is built from augmented variants of a single reference image rather than from a large dataset, exploiting the fixed-camera assumption: the camera does not move, so the background is stable across days and the only source of variation is illumination. Photometric augmentation at build time teaches the memory to tolerate this variation, making the decision threshold robust to normal fluctuations without requiring multiple observation days.

It is worth stressing why the alternative - populating the memory with a collection of *similar* exit scenes (many different corridors, many different doors) - would be both infeasible and counterproductive. Such scenes are visually too heterogeneous:

a memory mixing dozens of unrelated corridors and doorways would model the generic appearance of “a corridor” or “a door”, which is precisely the wrong target. Its feature manifold would be so broad that an obstacle in any specific scene could fall close to some unrelated normal scene in the bank and be scored as normal, destroying the detector’s sensitivity. The task is not to recognise corridors and doors in general, but to recognise *this* particular exit as it normally looks. This is the natural consequence of the deployment model, in which each camera monitors exactly one exit route: the normal class for that camera is its own fixed scene, and the only relevant variability is the illumination and the background change of that scene over time. A per-camera memory built from one reference image is therefore not a limitation imposed by scarce data, but the correct modelling choice for the objective. The figure 3.1 shows all the pipeline steps.

Feature extraction

Let I_{ref} denote the normal reference image of a monitored location. Every image is resized to a fixed resolution and passed through a frozen convolutional backbone f . Following the original PatchCore formulation, features are read from two intermediate blocks, ℓ_2 and ℓ_3 : the shallower block preserves local detail while the deeper block carries semantic context, and their combination balances the two. The shallowest block encodes mostly low-level texture with little semantic content, and the deepest block is too abstract and biased toward the pre-training classes; both are therefore excluded.

The deeper feature map is upsampled bilinearly to the spatial resolution of the shallower one and concatenated along the channel axis, yielding a joint tensor of patch vectors arranged on a $h \times w$ grid. A local average pooling over a 3×3 neighbourhood (the neighbourhood aggregation of [16]) is then applied: each patch vector is replaced by the mean of its 3×3 spatial neighbourhood,

$$\phi(x) = \frac{1}{|\mathcal{N}_3(x)|} \sum_{x' \in \mathcal{N}_3(x)} \tilde{\phi}(x'), \quad (3.1)$$

where $\tilde{\phi}(x')$ is the raw concatenated feature at grid location x' and $\mathcal{N}_3(x)$ is the 3×3 window centred at x . This enlarges the receptive field of each vector and reduces sensitivity to pixel-level noise. The result is a set of $N_p = h \cdot w$ patch feature vectors per image.

Build phase

The build phase constructs the *memory bank* $\mathcal{M}$, the finite set of normal patch features against which every test patch will later be compared. It proceeds in three steps - augmentation, feature accumulation, and coreset reduction - each of which is detailed below. The guiding principle is that $\mathcal{M}$ must represent the full appearance of the normal scene under its expected illumination variation, yet remain small enough for a real-time nearest-neighbour search on an edge device.

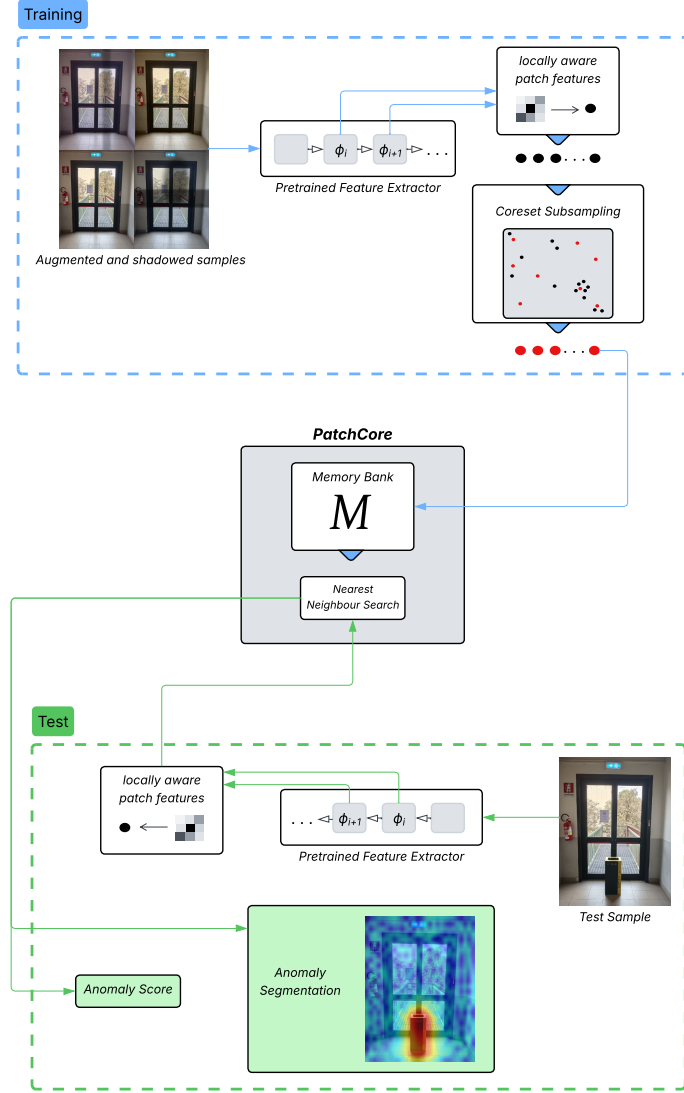


Figure 3.1: Schematic of the P3 pipeline. The memory bank is built from augmented variants of a single reference image, and test patches are scored by their distance to the nearest bank entry. The decision threshold is calibrated on additional augmented variants.

The build phase is a *one-off, offline* operation, performed once when the camera is installed and its reference frame I_{ref} is captured. The resulting memory bank is then stored and reused unchanged for every subsequent inference; nothing in the build phase runs at monitoring time. A rebuild is required only when the *normal* reference scene itself changes in a way that is not a transient occlusion - for example a permanent rearrangement of the furniture in view, a repainting, or a change of the camera's mounting - so that the old I_{ref} no longer describes the normal state. Routine illumination changes and background variation over time are deliberately *not* a reason to rebuild: they are exactly what the augmentation and

the calibration are designed to absorb. This separation between a rare, offline build and a continuous, lightweight inference is what makes the method practical on an edge device.

Step 1 - Augmentation. A single reference frame is too narrow a description of the normal scene: it captures one instant of illumination, whereas the same fixed camera will, over days, see the scene under brighter or dimmer light, warmer or cooler colour temperature, and varying sensor noise. To let the memory tolerate this variation, n_{aug} augmented variants are generated from I_{ref} , where n_{aug} is the number of bank variants. The augmentations are predominantly photometric - brightness, contrast, saturation, colour-temperature shifts, additive noise, and a mild blur - each motivated by a physical source of normal variation (time of day, light type, sensor noise, slight defocus). *Large* geometric transforms (flips, wide rotations, crops) are deliberately avoided: the camera is physically fixed, so orientation is semantically meaningful and must not be treated as a source of invariance, otherwise the memory would also accept an obstacle appearing in a flipped or arbitrarily relocated position as normal.

A controlled exception is a *small* geometric perturbation, off by default and enabled only for cameras subject to micro-vibration. When active, each variant is additionally shifted by a random translation of at most δ_{max} pixels on each axis and rotated by a random angle of at most ρ_{max} degrees, with δ_{max} of a few pixels and ρ_{max} of a couple of degrees. Its purpose is opposite to that of the forbidden large transforms: rather than making the memory invariant to where an obstacle sits, it makes the memory tolerant to a slight global displacement of the *whole* fixed scene, as produced by mechanical vibration or a small mounting drift. Such a displacement moves every patch by the same tiny amount, so a few-pixel jitter keeps a normal scene normal without ever spanning the distance an obstacle would cover. This is the cheapest of the available robustness measures, since it acts entirely at build time and adds no inference-time cost.

Step 2 - Feature accumulation. Each of the n_{aug} variants is passed through the feature pipeline of equation (3.1), producing N_p patch vectors per variant. Stacking the patches of all variants gives the *raw memory*

$$\mathcal{M}_{\text{raw}} = \bigcup_{v=1}^{n_{\text{aug}}} \{ \phi^{(v)}(x) : x \in \text{grid} \}, \quad |\mathcal{M}_{\text{raw}}| = n_{\text{aug}} \cdot N_p, \quad (3.2)$$

where $\phi^{(v)}(x)$ is the aggregated feature at grid location x of variant v . This mirrors the memory construction of [16], $\mathcal{M} = \bigcup_{x_i \in \mathcal{X}_N} \mathcal{P}(\phi(x_i))$, with the nominal training set $\mathcal{X}_N$ here instantiated by the augmented variants of the single reference I_{ref} . Each spatial location of the scene thus contributes n_{aug} nearby vectors - one per illumination variant - which together trace out the small region of feature space that a normal patch at that location can occupy. This is exactly the tolerance the

detector needs: at test time a normal patch, under any illumination within the augmented range, will fall close to at least one of these vectors and receive a low anomaly score.

Step 3 - Coreset reduction. The raw memory is highly redundant: adjacent grid locations on flat surfaces (a wall, a floor, a door panel) and the n_{aug} illumination copies of each location produce many near-duplicate vectors. Storing all of them would inflate both the memory footprint and the nearest-neighbour cost without improving coverage. The memory is therefore compressed to a coreset $\mathcal{M} \subset \mathcal{M}_{\text{raw}}$ of size a fraction p of the original - the *coreset ratio* - that minimises the worst case distance of any raw vector to its closest retained vector [18, 16]:

$$\mathcal{M}^* = \arg \min_{\mathcal{M} \subset \mathcal{M}_{\text{raw}}} \max_{m \in \mathcal{M}_{\text{raw}}} \min_{m' \in \mathcal{M}} \|m - m'\|_2. \quad (3.3)$$

This minimax (facility-location) objective is NP-hard, so it is approximated by a greedy farthest-point procedure: starting from one random point, the algorithm repeatedly adds the still unselected vector that is *farthest* from the current selection,

$$m_t = \arg \max_{m \in \mathcal{M}_{\text{raw}} \setminus \mathcal{M}_t} \min_{m' \in \mathcal{M}_t} \|m - m'\|_2, \quad \mathcal{M}_{t+1} = \mathcal{M}_t \cup \{m_t\}. \quad (3.4)$$

Choosing, at each step, the point that maximises its minimum distance to the already-selected set spreads the retained vectors as evenly as possible over the occupied region of feature space: dense clusters of near-duplicates are summarised by a few representatives, while isolated points - which carry unique information about the normal scene - are kept. The procedure stops when $|\mathcal{M}| = \lceil p \cdot |\mathcal{M}_{\text{raw}}| \rceil$, subject to a small floor on the number of retained points so that even a very low p leaves a usable bank. Because evaluating equation (3.4) directly in the full feature dimension is expensive, the distance comparisons are performed on a low-dimensional random projection ψ that approximately preserves pairwise Euclidean distances (a Johnson-Lindenstrauss embedding [21]), exactly as in [16], while the selected vectors are stored at their original dimension. The result is a compact bank that fits in a few megabytes and enables sub-millisecond nearest-neighbour lookups, a prerequisite for edge deployment, while retaining near-uniform coverage of the normal-feature manifold.

Threshold calibration

Design space. Several threshold strategies are possible for a one-class detector: a fixed global value, a per-camera percentile, a ROC-optimal cut calibrated on labelled anomalies, or a parametric rule derived from the distribution of normal scores. A fixed global threshold requires that the absolute scale of anomaly scores be consistent across cameras, which does not hold in practice - a plain opaque door scores

structurally lower than a glass door with a visible outdoor background, so a threshold tuned on one would misfire on the other. A ROC-optimal threshold requires labelled anomalies, violating the one-class premise and risking overfitting to the appearance of the synthetic anomalies used for tuning. A per-camera percentile is more sensitive to outliers in the small calibration sample. The chosen strategy is a parametric rule that adapts to each scene using only normal data.

Parametric rule. For each of n_{cal} additional augmented variants of I_{ref} (the *calibration variants*, disjoint from the bank variants and generated with an independent random seed), the maximum patch anomaly score over the image is computed. These n_{cal} scores form the empirical distribution of the maximum anomaly score of a normal scene, with mean μ and standard deviation σ . The decision threshold is

$$\theta = \mu + k \sigma, \quad (3.5)$$

where k is a sensitivity hyperparameter. This is the rule of statistical process control (Shewhart charts): an event is flagged when it lies k standard deviations above the normal mean. Under a Gaussian assumption, $k = 3$ corresponds to a theoretical false-positive rate of about 0.13% and $k = 4$ to about 0.003%; the calibration scores are not exactly Gaussian (they are maxima of correlated patch scores), but k still provides a monotone, interpretable control over the false-positive/recall trade-off. It is the sole deployment-time hyperparameter and encodes the asymmetric cost of a missed obstruction relative to a false alarm: for safety-critical exits k is set low (2.5 to 3.0) to favour recall, and higher (4.0 to 5.0) where false alarms are operationally costly.

Coverage dependency. Because μ and σ are estimated on augmented variants, they reflect only the score variability those augmentations induce. If a real source of variation (cast shadows, light beams, outdoor background changes through glass) is absent from the augmentation scheme, σ is underestimated and real test scores under that variation exceed θ more often than expected. This dependency is the root cause of the shadow false-positive problem (Section 5.1) and motivates the shadow and light augmentation introduced below.

Test phase and anomaly map

At inference, the query image is processed by the same feature pipeline to produce its patch vectors. Each patch vector ϕ is compared to the memory bank through nearest-neighbour search; let m^* be its nearest bank neighbour, so that the raw patch distance is $s^*(\phi) = \|\phi - m^*\|_2$. Let $\mathcal{N}_b(m^*)$ be the set of b nearest neighbours of m^* within the bank. Following equation 7 of [16], the patch score is re-weighted as

$$s(\phi) = w(\phi) s^*(\phi), \quad w(\phi) = 1 - \frac{\exp\|\phi - m^*\|}{\sum_{m \in \mathcal{N}_b(m^*)} \exp\|\phi - m\|}. \quad (3.6)$$

Two points deserve care. First, the distances in the denominator are between the test patch ϕ and the members of $\mathcal{N}_b(m^*)$, *not* between m^* and its own neighbours - a detail required for fidelity to the original formulation. Second, because m^* is the nearest neighbour, every distance in the denominator is at least $\|\phi - m^*\|$, so the fraction is at most $1/b$ and the weight is bounded as $w(\phi) \in [1 - 1/b, 1)$. The re-weighting therefore never inflates a patch score above its raw distance s^* : it leaves $s(\phi)$ close to s^* when m^* is isolated (its neighbours are far, the denominator is large, $w \rightarrow 1$), and discounts it by up to a factor $1 - 1/b$ when m^* sits in a dense region of common normal patterns (its neighbours are close, $w \rightarrow 1 - 1/b$). In relative terms this promotes patches matched to rare normal prototypes - the configuration most likely to indicate a genuine anomaly.

The original paper assigns the *image*-level score to the single most anomalous patch, $\arg \max_{\phi} s^*(\phi)$, re-weighting only that one (equations 6-7 of [16]). The present work instead evaluates equation (3.6) for *every* patch, arranges the scores on the spatial grid to form an anomaly map, upsamples it bilinearly, and smooths it with a Gaussian filter; the scalar image score is then the maximum of this smoothed map, $s_{\max}$, and the binary decision is $s_{\max} > \theta$. This is a deliberate variant of the paper’s single-patch score, adopted for two reasons: the full per-patch map is required by the spatial gating of Section 3.2.1, which operates on connected components of the map; and taking the maximum of the smoothed map - rather than of a single raw patch - is more robust to an isolated noisy patch, since smoothing broadens genuine anomaly blobs (helping small objects that span few patches) while attenuating one-pixel spikes. The two formulations coincide when no smoothing is applied and the maximiser of $w(\phi) s^*(\phi)$ is also the maximiser of s^* .

Shadow and light augmentation

The photometric-only memory contains no variants with cast shadows or localised light excess. This leaves a coverage gap: at test time, a shadow projected on the floor by a passer-by outside the field of view, or a light beam from a skylight or a glass door, produces patch features distant from all bank entries and trips the threshold even though no obstruction is present. The two disturbances are symmetric - darkening versus brightening - and are addressed symmetrically by adding synthetic variants of each kind to the augmentation scheme. Three modalities are modelled for each (elliptical, directional, stripe), covering the physically plausible patterns in corridor and doorway scenes; all borders are blurred to avoid unrealistic high-frequency edges.

A disturbance is added to a build variant with probability p_{sh} (shadow) or p_{li} (light), and to a calibration variant with probability $p_{\text{sh}}^{\text{cal}}$ and $p_{\text{li}}^{\text{cal}}$ respectively. A central design choice is to *decouple* the build and calibration probabilities, because the two phases have opposing requirements:

- the **memory** needs maximum disturbance coverage so that test-time shadow/-light patches find near neighbours and produce low distances - hence a high

build probability ($p_{\text{sh}} = p_{\text{li}} = 0.4$);

- the **calibration** needs a stable score distribution to yield a reliable μ and σ - hence a low calibration probability ($p_{\text{sh}}^{\text{cal}} = p_{\text{li}}^{\text{cal}} = 0.1$).

Setting the calibration probability equal to the build probability makes σ high-variance: a few unlucky calibration variants with heavy disturbances inflate σ , and the $k\sigma$ amplification produces per-camera thresholds that are either too high (false negatives) or too low (false positives). Decoupling removes this instability. The two families are independent: a variant may receive a shadow, a light, both, or neither. The disturbances are deliberately kept out of the test images used to measure robustness, so that the reported false-positive rates reflect generalisation to disturbances never seen at build time.

Spatial gating via annotated polygons

Glass doors introduce a distinct failure mode: the transparent panel makes the external background visible, and any movement or lighting change beyond the door generates anomaly blobs in the glass region without any physical obstruction. The principled remedy exploits a physical-grounding constraint: *any real obstruction rests on the floor and therefore intersects the door threshold*.

Door-frame polygons (left jamb, right jamb, threshold, architrave) are annotated once on the reference image and rasterised into a binary *frame mask* F (Figure 3.2). The gating rule operates on the connected components of the thresholded anomaly map: the map is binarised at θ , its 8-connected components are labelled, and a component C is accepted only if

$$|C \cap F| \geq \tau, \quad (3.7)$$

where τ is the minimum overlap, in pixels, between a component and the frame mask. The final score is the maximum of the continuous anomaly map over the accepted components, or zero if none survives. A component confined to the glass panel has zero overlap with F and is rejected, whereas a floor-grounded obstruction produces a component reaching the threshold or jambs and is accepted. The same rule applies to opaque doors when the panel exhibits strong illumination variation: specular highlights stay on the panel, while grounded obstructions reach the frame.

Structural limitations. Gating assumes the frame geometry is stable and visible at test time. An *open door* violates this: the threshold and jambs change radically, producing an out-of-distribution component inside F (a structural false positive), while an obstacle in the open doorway no longer intersects the misplaced mask (a structural false negative). A thin-stemmed grounded object (a floor fan, a sign on a pole) has only a narrow contact with the frame that may fail the overlap criterion τ . These cases share one root: reachability of the frame mask by a genuine anomaly component depends on frame visibility, a property of the scene rather than of the



Figure 3.2: Schematic of the spatial gating rule. A component C is accepted if it overlaps the frame mask F by at least τ pixels, otherwise it is rejected.

model. They are declared as limitations of the method; the system is designed and validated for the nominal closed-door condition, dominant in continuous surveillance.

Calibration stabilisation: pooled sigma

The per-camera rule $\theta_i = \mu_i + k\sigma_i$ has a structural instability. The per-camera standard deviation σ_i is estimated from a small calibration sample and carries a relative error on the order of 18%; this error propagates one-to-one into the $k\sigma_i$ term, and across many cameras it produces a long-tailed distribution of thresholds - a few cameras receive thresholds high enough to miss all obstructions, others low enough to generate shadow false positives, even though the underlying scenes are not fundamentally different. The extreme σ_i values are stochastic calibration outliers, not scene outliers: the same camera re-calibrated with a different seed would receive a normal threshold.

The fix replaces the per-camera σ_i with a single population estimate while keeping μ_i local:

$$\theta_i = \mu_i + k\sigma_{\text{pop}}, \quad \sigma_{\text{pop}} = \text{median}(\sigma_1, \dots, \sigma_N), \quad (3.8)$$

where N is the number of cameras. The median is robust to the outlier σ_i that caused the tail. Keeping μ_i per-camera preserves scene-specific adaptation (the absolute score level of each scene), while pooling σ removes the noisy multiplicative

term. Under the per-camera rule $\text{Var}(\theta_i) = \text{Var}(\mu_i) + k^2 \text{Var}(\sigma_i) + 2k \text{Cov}(\mu_i, \sigma_i)$, whereas under the pooled rule it collapses to $\text{Var}(\theta_i) = \text{Var}(\mu_i)$, i.e. to the intrinsic scene variability alone.

Pooled sigma is, by construction, a shrinkage (regularisation) estimator: it trades a small bias - forcing each camera toward the population median - for a large variance reduction. In the data-starved regime of this thesis (a one-shot memory from a single image with sparse synthetic calibration) the per-camera σ_i is noise-dominated and the true cross-camera heterogeneity is near zero, so full pooling is the correct corner. The principled generalisation is *partial pooling*, $\sigma_i^* = (1 - w_i) \sigma_{\text{pop}} + w_i \sigma_i$, with the weight w_i rising as a camera accumulates real normal frames; full pooling ($w_i = 0$) is the cold-start corner and remains valuable for newly installed cameras.

3.3 Dataset Description

[Description of the dataset used to build, calibrate, and evaluate the pipeline - to be completed.]

Chapter 4

Experimental Setup

4.1 Pipeline Implementation

4.1.1 Unsupervised Anomaly Detection with PatchCore (P3)

This section reports the concrete technical settings of the method derived in Chapter 3. Each monitored exit is handled by an independent memory bank built from a single reference frame, reflecting the deployment reality of a fixed camera with a stable normal scene.

Backbone and feature extraction

The backbone is a WideResNet50-2 [22] pre-trained on ImageNet and used frozen (no fine-tuning). Every image is resized to 224×224 and normalised with ImageNet statistics. Features are read from blocks ℓ_2 and ℓ_3 :

Block	Spatial resolution	Channels
ℓ_2	28×28	512
ℓ_3	14×14	1024

Table 4.1: Feature maps read from the frozen backbone at 224×224 input.

The ℓ_3 map is upsampled bilinearly to 28×28 and concatenated with ℓ_2 along the channel axis, giving a tensor of shape $[1536, 28, 28]$. The 3×3 neighbourhood average pooling of equation (3.1) is applied with unit stride and same-padding, and the result is flattened to $N_p = 784$ patch vectors of dimension 1536 per image. The shallowest block (56×56 , low-level texture) and the deepest block (7×7 , class-biased abstraction) are not used.

Build and calibration settings

The bank is built from $n_{\text{aug}} = 15$ photometric variants of I_{ref} (random seed 42); the original reference is excluded. The photometric augmentation ranges are listed in

Table 4.2.

Transform	Range	Rationale
Brightness	$\times [0.85, 1.15]$	Morning / evening / artificial light
Contrast	$\times [0.90, 1.10]$	Direct vs. diffuse illumination
Saturation	$\times [0.90, 1.10]$	LED / fluorescent / incandescent
Red-channel shift	$[-8, +8]$	Colour temperature
Blue-channel shift	$[-20, +20]$	Colour temperature
Gaussian noise	$\sigma \in [1, 4]$	Sensor noise
Gaussian blur	$r \in [0.5, 1.5], p = 0.3$	Slight lens defocus

Table 4.2: Photometric augmentation applied at build time. No large geometric transform is used, as the camera is fixed.

In addition to the photometric transforms, an optional small geometric perturbation is available for cameras subject to micro-vibration, disabled by default ($\delta_{\max} = 0$, $\rho_{\max} = 0$). When enabled, every variant receives a per-variant random translation drawn uniformly in $[-\delta_{\max}, +\delta_{\max}]$ pixels on each axis and a random rotation drawn uniformly in $[-\rho_{\max}, +\rho_{\max}]$ degrees, applied by bilinear affine warping at the original resolution; typical settings are $\delta_{\max} = 8$ px and $\rho_{\max} = 2^\circ$. Unlike the forbidden flips and wide rotations, this jitter perturbs the whole scene rigidly and teaches the memory to tolerate a small global displacement at no inference-time cost. It is left off for all experiments reported in Chapter 5, where the data-collection protocol (fixed foot position, braced camera) keeps hand-held variation within the native tolerance of the neighbourhood pooling and the anomaly-map smoothing.

The raw memory of $n_{\text{aug}} \cdot N_p = 15 \times 784 = 11760$ vectors is reduced by greedy coresets selection to a fraction $p = 0.01$ (retaining at least 50 points), leaving about 117 vectors. For efficiency the greedy selection of equation (3.4) runs on a sparse random projection to 128 dimensions (a Johnson-Lindenstrauss embedding), while the final coreset is taken from the original 1536-dimensional features.

The threshold is calibrated on $n_{\text{cal}} = 10$ further variants generated with seed 1000, disjoint from the bank seed to guarantee statistical independence; later experiments raise this to $n_{\text{cal}} = 30$ to reduce the sampling error on σ . The re-weighting of equation (3.6) uses a neighbourhood of size $b = 9$, and the anomaly map is smoothed with a Gaussian filter of $\sigma = 4$. The memory bank is serialised to a single file holding the coreset features together with the calibration parameters μ and σ .

Anti-leakage separation

The bank, calibration, and test sets are strictly disjoint (Table 4.3): no data used to build the memory or calibrate the threshold enters the test phase, and I_{ref} is reserved as the only genuinely unseen normal sample.

Set	Seed	Contains I_{ref} ?	Role
Bank variants ($n_{\text{aug}} = 15$)	42	No	Builds the feature memory
Calibration variants ($n_{\text{cal}} = 10$)	1000	No	Estimates μ and σ
Normal test	–	Yes	Independent normal sample
Obstructed test	–	–	Anomalous query

Table 4.3: Disjoint partition of data across build, calibration, and test.

Shadow and light augmentation

Both disturbance families implement three modalities, summarised in Table 4.4. They are applied after the photometric transforms, stacking $k_s \in [1, 3]$ instances per affected variant. The light family adds two axes absent from shadows: a diagonal (solar) stripe parameterised by a continuous incidence angle, and a warm colour shift (red up, blue down) on the elliptical and diagonal-stripe lights, modelling the chromaticity of direct sunlight. As a consequence the light subspace is intrinsically richer, so the raw values of light and shadow false-positive rates are not directly comparable; only within-category comparisons (across configurations) are legitimate.

Modality	Shadow	Light
Elliptical	dark ellipse, blurred border	bright ellipse (beam, specular)
Directional	dark edge gradient	bright edge gradient (lateral window)
Stripe	dark band (2 orientations)	bright band (3 orient., solar diagonal)
Colour shift	none	warm tint on elliptical / diagonal

Table 4.4: Synthetic disturbance modalities. Build probabilities $p_{\text{sh}}, p_{\text{li}} = 0.4$; calibration probabilities $p_{\text{sh}}^{\text{cal}}, p_{\text{li}}^{\text{cal}} = 0.1$.

To formally separate the test-time disturbances from the build-time ones, the build augmentation draws its shadows and lights from a dedicated random generator seeded disjointly from the generator that produces the test disturbances, so the two sets never coincide.

ROI annotation and connected-component gating

The frame mask of equation (3.7) is produced by a three-stage tool chain. A lightweight interactive annotator draws one polygon per frame element (left jamb, right jamb, central mullion, threshold, architrave, or a catch-all label) and writes one JSON file per image, storing polygons in original-image coordinates to avoid precision loss after resizing. An importer loads these polygons into the database, matching by relative file path. At evaluation time the polygon set of the reference frame is rasterised to the original resolution, the smoothed anomaly map is binarised at θ_i , its 8-connected components are extracted, and a component is kept only if its overlap with the frame mask reaches the minimum $\tau = 10$ pixels. The score is

the maximum of the continuous map over the surviving components, or zero if none survive.

The binarisation threshold used to form the components is the same per-camera θ_i used for the decision, so the sensitivity k acts at two stages - which pixels survive binarisation, and which final scores clear the decision. A legacy rectangular ROI can be composed in front of the polygon gating: the rectangle zeroes the map outside the door bounding box first (suppressing floor reflections and solar shadows), and the polygon rule then filters the remaining glass-panel variation. The full image is always included in the memory bank: excluding the glass region would leave its patches without any neighbour at test time, inflating their distance by mere absence of reference and making the panel anomalous by construction.

Pooled-sigma evaluation

The pooled threshold of equation (3.8) is computed in two passes over the cameras. The first pass builds each bank, calibrates μ_i and σ_i , and scores all test samples without yet deciding their label. The second pass collects the σ_i , sets σ_{pop} to their median, and recomputes for every sample the threshold $\theta_i = \mu_i + k \sigma_{\text{pop}}$, the normalised score $(s - \mu_i)/\sigma_{\text{pop}}$, and the binary decision. The normalisation makes scores comparable across cameras of different absolute scale, so that the aggregated AUROC measures intrinsic separability rather than scale differences. The mode (per-camera or pooled) and the value of σ_{pop} are recorded with the experiment for reproducibility. Pooling affects neither inference time nor memory size, since calibration variants never enter the coreset; its only cost is a one-off computation at build time.

4.2 Application

[General description of the application, its modules, and overall architecture - to be completed.]

Chapter 5

Experimental Analysis

5.1 Pipeline Results

5.1.1 Unsupervised Anomaly Detection with PatchCore (P3)

Evaluation protocol

All results are obtained through a database-driven evaluation that, for each normal reference frame, (i) builds the memory bank from augmented variants of the reference (excluding the original), (ii) calibrates the threshold on independent variants, (iii) scores the original reference as the normal test sample, and (iv) scores the obstructed frames linked to that reference. Scores are normalised per camera as $(s - \mu_i)/\sigma_i$ before aggregating the AUROC, while threshold-based metrics (false-positive rate FPR, true-positive rate TPR, and F_1) use the per-camera threshold θ_i so that each binary decision reflects its own calibration. Unless stated otherwise, experiments run on the copy-paste synthetic dataset (split *test*, doors and corridors together), about 990 reference frames each with one or more synthetic obstructed variants.

The evaluation is organised as a progressive ablation, in which each component is introduced in response to a failure mode identified empirically (Table 5.1).

Phase	Configuration	Purpose
A	Baseline (no disturbances, no ROI)	Characterise base behaviour
B	Baseline + shadow test set	Quantify shadow FPR
C	Shadow augmentation + pooled σ	Fix shadow FPR, stabilise θ
D	Real doors + polygon gating	Validate full pipeline, recalibrate k
E	Light augmentation ablation	Reduce light FPR; separate coverage from budget

Table 5.1: Progressive-ablation structure of the evaluation.

Phase A: baseline on the copy-paste dataset

The baseline ($n_{\text{aug}} = 15$, $n_{\text{cal}} = 10$, no disturbance augmentation, $k = 3$) cleanly separates normal from obstructed scenes (Table 5.2). The near-perfect AUROC and the gap of several standard deviations between the class means confirm that obstacle patches fall well outside the distribution of normal door and corridor patches. The residual 0.5% FPR corresponds to about five false alarms over 990 frames. This establishes the detector as viable and validates the evaluation methodology.

Category	Mean score	Std	Rate
Normal	2.309	0.156	FPR 0.5%
Obstructed	4.648	0.348	TPR 100%
AUROC	≈ 1.000		

Table 5.2: Phase A baseline on the synthetic test set.

Phase B: shadow vulnerability

A test set of shadow-augmented normal scenes (one synthetic variant per reference, $k_s \in [1, 3]$ stacked shadows) is scored against the baseline model. These are normal scenes with no obstruction, yet 90.1% are flagged as anomalous (Table 5.3). Shadow patches - dark, smoothly-bordered regions - produce features unlike any bank entry, which contains only shadow-free photometric variants; the baseline cannot distinguish a cast shadow from a physical obstruction at the feature level.

Category	Mean	Std	Range	Rate
Normal	2.309	0.156	[1.87, 3.13]	FPR 0.5%
Shadow normal	3.189	0.495	[2.10, 4.51]	FPR 90.1%
Obstructed	4.648	0.348	[3.54, 5.87]	TPR 100%

Table 5.3: Phase B: shadow-normal scores overlap the obstructed range [3.54, 4.51].

The distributions overlap in [3.54, 4.51], so no fixed global threshold achieves acceptable shadow FPR and TPR simultaneously: at a cut of 4.00 the shadow FPR is 4.8% but TPR has already dropped to 97.3%, and pushing the cut to 4.50 to reach 0.1% shadow FPR collapses TPR to 67.4%. The problem is not threshold calibration but a distribution mismatch between the shadow-free memory and the shadow-bearing test set.

Phase C: shadow augmentation and pooled sigma

Introducing shadow augmentation in the build phase, then decoupling and pooling the calibration, traces a clear path (Table 5.4). Coupling the build and calibration probabilities ($p_{\text{sh}} = p_{\text{sh}}^{\text{cal}} = 0.4$) reduces shadow FPR by two orders of magnitude but over-inflates the threshold, dropping TPR to 70.6%; the cause is a long tail in

the per-camera threshold distribution, where 12% of cameras generate 86% of all false negatives. Decoupling ($p_{\text{sh}}^{\text{cal}} = 0.1$, $n_{\text{cal}} = 30$) recovers TPR but swings to an over-permissive operating point. Replacing the per-camera σ_i with the pooled median $\sigma_{\text{pop}} = 0.300$ at $k = 4$ eliminates the residual threshold variance and reaches the best operating point.

Configuration	Clean FPR	Shadow FPR	TPR	F_1	σ_θ
Baseline (no aug)	0.5%	90.1%	100%	0.688	0.158
Coupled ($k = 3$)	0.0%	2.1%	70.6%	0.818	0.539
Decoupled ($k = 3$)	0.0%	28.4%	99.4%	0.873	0.388
Decoupled ($k = 4$)	0.0%	13.5%	93.4%	0.903	0.497
Pooled σ ($k = 4$)	0.0%	12.2%	99.8%	0.941	0.137

Table 5.4: Phase C: shadow augmentation and calibration stabilisation. σ_θ is the standard deviation of the per-camera thresholds.

Pooling collapses the threshold range from [2.28, 6.09] to [3.24, 4.16] (a 73% reduction in σ_θ) and removes the pathological tail entirely: no camera exceeds $\theta = 4.16$ and TPR stays at 99.3% or higher in every threshold bucket. The 3.8 percentage-point F_1 gain is due entirely to variance reduction, not to any change in the backbone, whose AUROC stays at 0.998. The choice $k = 4$ follows a conservative four-sigma rule and was not tuned on the test set: the F_1 -optimal value is about 3.9, but reporting it would constitute test-set tuning.

Phase D: real-world validation on real doors

Phase D is the first evaluation on real photographs rather than copy-paste composites. The test set consists of six real doors photographed on site (a mix of opaque and glass doors), each with one annotated reference frame, comprising 30 real obstructed images, 30 unobstructed images, and the six references. Consistent with the fixed-camera assumption, all shots of a door share the same framing, so the negatives measure the FPR under normal content and illumination variability at fixed geometry. The full pipeline is active: pooled σ plus connected-component gating ($\tau = 10$).

Validated on the synthetic suites, $k = 4$ proved too conservative on real scenes; lowering it to $k = 3$ raises recall at no FPR cost (Table 5.5). Because the pooled median(σ) = 0.213 is a shared constant, lowering k shifts every threshold down by the same amount without widening the distribution - the per-camera variance amplification of the naive rule does not reappear.

Comparing the two runs separates the false negatives by cause. Two are *calibration-limited* and are recovered at $k = 3$ through the two distinct stages where k acts: one is a decision-stage miss (a valid gated component scored just below the decision threshold at $k = 4$), the other a binarisation-stage miss (the obstruction’s anomaly peak fell just below the local binarisation threshold at $k = 4$, leaving an empty bi-

Metric	$k = 4$	$k = 3$	Δ
Recall (TPR)	83.3%	90.0%	+6.7 pp
FPR	3.3%	3.3%	invariant
Precision	96.2%	96.4%	+0.2 pp
F_1	0.893	0.931	+0.038
Accuracy	90.0%	93.3%	+3.3 pp

Table 5.5: Phase D: recalibration of k on the real test set.

nary map). The latter shows that θ controls not only the decision but which pixels exist for gating to evaluate, so an overly high k can erase a low-contrast obstruction before gating sees it. Three remain zeroed even at $k = 3$ and are *gating-limited*: the prototypical case is a free-standing floor fan whose wide head sits on the door panel (excluded from the frame mask by design) and whose only frame contact, the thin pole, narrows to 1-2 px after resizing and fails the $\tau = 10$ overlap. PatchCore detects the fan correctly; the failure is purely geometric, confirming the structural limitation anticipated in Chapter 3: the grounded-obstacle assumption holds for wide-profile objects (carts, wheelchairs, bins) but not for thin-stemmed ones (fans, coat racks, sign posts). A single FP is a glass door whose background changed behind the panel. With order-of-tens samples, these numbers are a qualitative validation of the complete pipeline, not a production benchmark.

Phase E: light augmentation ablation

Phase E addresses the symmetric counterpart of the shadow failure mode - localised brightness excess - evaluated on a light-augmented test set (≈ 1100 normal scenes) at $k = 4$, pooled σ , $n_{\text{cal}} = 30$. Five configurations are compared (Table 5.6), where the *budget* $p_{\text{sh}} + p_{\text{li}}$ is the expected disturbance probability per variant, a proxy for total augmentation intensity. A fifth fair-budget control (L4) disentangles two effects that L2/L3 confound: increased coverage (adding the light type) and increased total intensity.

Config	p_{sh}	p_{li}	Budget	Shadow FPR	Light FPR	F_1
L0 shadow-only	0.4	0.0	0.40	12.2%	5.8%	0.917
L1 light-only	0.0	0.4	0.40	37.7%	23.2%	0.767
L2 SL-15	0.4	0.4	0.80	2.2%	0.30%	0.982
L3 SL-25	0.4	0.4	0.80	1.1%	0.10%	0.982
L4 fair-budget	0.23	0.23	0.46	11.2%	4.6%	0.929

Table 5.6: Phase E light-augmentation ablation. TPR is $\geq 97.7\%$ throughout. L0/L2 use $n_{\text{aug}} = 15$, L3 uses $n_{\text{aug}} = 25$.

Four findings emerge. First, *light is a milder disturbance than shadow*: even with no light coverage (L0) the light FPR is 5.8%, below the 12.2% shadow FPR, because

a light beam pushes features less out of distribution than a cast shadow. Second, *adding light augmentation works and helps shadows too*: from L0 to L2 the light FPR falls about nineteen-fold while the shadow FPR also improves rather than degrading. Third, L1 confirms that shadow augmentation is *structural, not incidental*: removing it collapses the per-camera threshold and inflates all FPRs, light included - light augmentation is a complement to shadow augmentation, not a substitute. Fourth, the fair-budget control shows that *the headline gain is mostly a budget effect*: L2/L3 apply twice the augmentation of L0, and the mean threshold grows monotonically with budget, tracking the FP reduction; at equal budget (L4 vs. L0) diversification alone yields only a small, cost-free improvement. The recommended configuration is SL-15 ($F_1 = 0.982$, light FPR 0.30%, shadow FPR 2.2%, recall 99.0%); SL-25 halves the FPRs again but trades recall for identical F_1 at a markedly longer build time, justified only when minimal FP is required. As established in Chapter 4, light and shadow FPR are not directly comparable; only within-category readings are valid.

Discrimination versus calibration

A cross-cutting question determines what the pipeline can legitimately claim: do the interventions improve the model’s *discriminative capacity* or only the *placement of the threshold*? The clean metric is the per-camera AUROC - the probability that an obstruction scores higher than a normal variant of the *same* camera, on raw scores, averaged over cameras. Within a camera μ_i and σ_i are constant, so raw and normalised scores differ by an affine transform that preserves ranking; the per-camera AUROC is therefore invariant to both threshold and normalisation and isolates intrinsic discrimination.

Config	raw AUROC	per-cam AUROC (shadow)	shadow FPR	TPR	F_1
no-aug	0.9969	0.996	90.1%	100%	0.688
shadow04	0.9981	0.996	2.1%	70.6%	0.818
decoupled	0.9981	0.996	28.4%	99.4%	0.873
decoupled $k = 4$	0.9981	0.996	13.5%	93.4%	0.903
pooled	0.9981	0.996	12.2%	99.8%	0.941

Table 5.7: Controlled comparison: shadow FPR swings 40× while raw and per-camera AUROC stay fixed.

The per-camera AUROC is saturated and nearly invariant to augmentation (Table 5.7): clean separation is perfect everywhere, the shadow value stays at 0.996 across the whole path - even with no shadow augmentation - and light improves only from 0.997 to 1.000. The shadow FPR meanwhile swings from 2.1% to 90% to 12% while the raw AUROC is fixed at 0.9981. The mechanism is a *distributional shift, not a re-ordering*: without shadows in the memory, shadow patches have no neighbours and score high, staying below the obstructed scores (ranking preserved) but above a threshold calibrated on clean scenes (FPR 90%); with shadows in the

memory, their scores drop below the threshold without changing order relative to obstructions. Augmentation shifts the disturbed-normal distribution downward relative to a fixed threshold; pooled σ then stabilises where that threshold lands across cameras. Neither adds discriminative power - they convert an already-excellent latent discrimination into a usable single operating point.

This conclusion holds on a test set where discrimination is near-saturated by design (copy-paste obstructions on the same background: only the object is anomalous, a strong signal, AUROC ≈ 1). On harder data - real obstructions, variable backgrounds - discrimination would have headroom and augmentation might contribute to separability as well; the per-camera AUROC argument is a claim about this dataset's regime and is stated as such.

Discussion

The progressive ablation shows that the failure modes of the method are diagnosable and fixable within the PatchCore framework without architectural changes. Cast shadows are addressed by synthetic shadow variants in the memory with decoupled calibration (shadow FPR 90.1% $\rightarrow$ 2.2% at SL-15); localised brightness excess is addressed symmetrically by light variants; glass-door backgrounds are addressed by connected-component gating exploiting the floor-grounding constraint; and threshold instability across cameras is addressed by pooling the σ estimate. The unifying finding is that none of these fixes change the backbone's discriminative capacity, which is saturated; the bottleneck is placing a single threshold robust to disturbances.

The conclusions carry two tiers of statistical weight. The synthetic phases ($N \approx 1000$) are quantitative validations with adequate power: the measured deltas of shadow augmentation, decoupling, and pooled σ are stable and generalisable to similar synthetic scenes. The real-door phase ($N \approx \text{tens}$) is a qualitative validation of the complete pipeline: it confirms the expected behaviour of gating plus pooled σ and motivates the k recalibration, but the absolute numbers are illustrative, not a production benchmark. The structural results - augmentation effectiveness, decoupling rationale, pooled-sigma variance reduction, and the discrimination-versus-calibration mechanism - are pipeline properties that generalise; the exact operating-point metrics should be recalibrated for any new deployment using a site-specific σ_{pop} and a k chosen from the application's cost ratio of false negatives to false positives. Larger-scale quantitative validation of the gating on real glass doors, conditioned on a semi-automatically annotated test set, is the natural next step.

5.2 Application Results

[Analysis of the end-to-end application results - to be completed.]

Chapter 6

Conclusions

[Summary and future work]

Appendix A

Questa è un'appendice

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Questa è un'altra sezione, ma non viene inserita nell'indice

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Bibliography

- [1] Paul Bergmann, Michael Fauser, David Sattlegger, and Carsten Steger. MVTec AD – a comprehensive real-world dataset for unsupervised anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9592–9600, 2019.
- [2] Yunkang Cao, Yuqi Cheng, Xiaohao Xu, Yiheng Zhang, Yihan Sun, Yuxiang Tan, Yuxin Zhang, Xiaonan Huang, and Weiming Shen. Visual anomaly detection under complex view-illumination interplay: A large-scale benchmark. arXiv preprint arXiv:2505.10996, 2025.
- [3] Dayou Chen, Long Chen, Yi Yang, Qiuchen Lu, Craig Hancock, Russell Lock, and Simon Sølvesten. A knowledge-driven vision-language reasoning approach for automated fire safety compliance checking in operational buildings. SSRN preprint, 2025. <https://ssrn.com/abstract=5387267>.
- [4] Niv Cohen and Yedid Hoshen. Sub-image anomaly detection with deep pyramid correspondences. arXiv preprint arXiv:2005.02357, 2020.
- [5] Thomas Defard, Aleksandr Setkov, Angélique Loesch, and Romaric Audigier. PaDiM: a patch distribution modeling framework for anomaly detection and localization. In *Pattern Recognition. ICPR International Workshops and Challenges*, pages 475–489, 2021.
- [6] Huu-Thanh Duong, Viet-Tuan Le, and Vinh Truong Hoang. Deep learning-based anomaly detection in video surveillance: A survey. *Sensors*, 23(11):5024, 2023.
- [7] M. Fikret Ercan and Asha Mathyalakan. Computer vision based safety inspection in public housing. In *TENCON 2024 – IEEE Region 10 Conference*, pages 1398–1401, 2024.
- [8] Gionatan Gallo, Francesco Di Rienzo, Federico Garzelli, Pietro Ducange, and Carlo Vallati. A smart system for personal protective equipment detection in industrial environments based on deep learning at the edge. *IEEE Access*, 10:110862–110878, 2022.

- [9] Shengtai Ju and Amy R. Reibman. Shadow augmentation for handwashing action recognition: from synthetic to real datasets. arXiv preprint arXiv:2410.03984, 2024.
- [10] Giustino Claudio Miglionico, Francesco Di Rienzo, Pietro Ducange, Francesco Marcelloni, and Carlo Vallati. A comparative analysis of models for real-time personal protective equipment detection on edge devices. In *2025 International Joint Conference on Neural Networks (IJCNN)*, 2025.
- [11] Giustino Claudio Miglionico, Pietro Ducange, Francesco Marcelloni, Carlo Vallati, and Francesco di Rienzo. Performance evaluation of YOLOv5 on edge devices for personal protective equipment detection. In *2024 International Joint Conference on Neural Networks (IJCNN)*, 2024.
- [12] Payal Mittal. A comprehensive survey of deep learning-based lightweight object detection models for edge devices. *Artificial Intelligence Review*, 57:242, 2024.
- [13] Zainab Ouairi, Sidi Ahmed Mahmoudi, and Mostapha Zbakh. Enhancing object detection in smart video surveillance: A survey of occlusion-handling approaches. *Electronics*, 13(3):541, 2024.
- [14] Devashree R. Patrikar and Mayur Rajaram Parate. Anomaly detection using edge computing in video surveillance system: review. *International Journal of Multimedia Information Retrieval*, 11:85–110, 2022.
- [15] Joseph Redmon, Santosh Divvala, Ross Girshick, and Ali Farhadi. You only look once: Unified, real-time object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 779–788, 2016.
- [16] Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter Gehler. Towards total recall in industrial anomaly detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14318–14328, 2022.
- [17] Marcos C. S. Santana, Leandro Aparecido Passos, Thierry P. Moreira, Danilo Colombo, Victor Hugo C. de Albuquerque, and João Paulo Papa. A novel siamese-based approach for scene change detection with applications to obstructed routes in hazardous environments. *IEEE Intelligent Systems*, 35(1):44–53, 2020.
- [18] Ozan Sener and Silvio Savarese. Active learning for convolutional neural networks: A core-set approach. In *International Conference on Learning Representations (ICLR)*, 2018.
- [19] Hyo-Ju Shin, Sang-Pil Jung, and Jin-Wook Kim. Improving object classification accuracy for detecting obstacles on indoor evacuation paths using YOLOv8. *KIEAE Journal*, 25(6):5–14, 2025.

- [20] Hamed Talebi, Nahid Khoshk Angabini, Arezoo Sarkheyli-Hägele, and Erdal Akin. Real time exit blockage detection via door-centric change monitoring for emergency evacuation. In *Proceedings of the 18th International Conference on Agents and Artificial Intelligence (ICAART)*, volume 3, pages 2648–2657, 2026.
- [21] Wikipedia contributors. Johnson–lindenstrauss lemma — Wikipedia, the free encyclopedia. https://en.wikipedia.org/w/index.php?title=Johnson%E2%80%93Lindenstrauss_lemma&oldid=1357719950, 2026. [Online; accessed 10-June-2026].
- [22] Sergey Zagoruyko and Nikos Komodakis. Wide residual networks. In *Proceedings of the British Machine Vision Conference (BMVC)*, 2016.
- [23] Syed Farhan Alam Zaidi, Jaehun Yang, Muhammad Sibtain Abbas, Rahat Hus-sain, Doyeop Lee, and Chansik Park. Vision-based construction safety monitoring utilizing temporal analysis to reduce false alarms. *Buildings*, 14(6):1878, 2024.